

Salience-Guided Question Generation: Feature Ablations, Methodological Insights, and Issue Analysis

Research Group Update

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NLP/ML Research Group

Agenda

- Problem framing and SQuAD context
- Pipeline and balanced prompt design
- Feature engineering and ablation setup
- Quantitative results and trade-offs
- Issue analysis from QA generation logs
- Limitations, future work, and discussion

Problem & Motivation

- SQuAD 1.1 has 100,000+ crowd-sourced QA pairs over 500+ Wikipedia articles.
- Answers are extractive spans; passages usually have 3–5 gold questions.
- In our setup: process train split after skipping first 2000 passages; generate 2 questions per passage.
- Standard QG often prioritizes fluency/diversity but can drift from key facts.
- Goal: use sentence salience to keep generation grounded in high-value passage content.

Proposed Approach

1. Sentence-level feature extraction.
2. Saliency scoring via classifier.
3. Select top-k salient sentences.
4. Balanced prompt conditions generation on salient evidence.
5. Generate 2 questions per passage.
6. Evaluate with ROUGE-L, BERTScore, QA-EM, QA-Consistency-F1, Answer-Recovery-F1.

Design principle: soft grounding, not hard constraints.

- Keep the model flexible and natural.
- Bias toward salient content for better answerability and consistency.

Salience Feature Engineering

Feature family	Rationale
Linguistic (surface, position, POS, lexical density, discourse markers)	Interpretable, cheap, structurally strong priors for answer-bearing sentences.
GPT-2 surprisal statistics (mean/sum/std/var/min/max)	Captures local uncertainty under causal LM; may mark information-rich regions.
BERT surprisal statistics (mean/sum/std/var/min/max)	Contextual uncertainty with bidirectional context; complementary signal.

Nuance: surprisal adds signal, but linguistic features remain surprisingly competitive.

Ablation Study Design (750 Passages)

Controlled setup

- Same balanced prompt, same sample size (750 passages), same generation count (1500 questions).
- Same seed and evaluation pipeline for comparability.

Conditions

- Full Features: linguistic + GPT-2 surprisal + BERT surprisal
- No Surprisal: linguistic-only
- Top-10 Features: compact high-signal subset

Why this design?

- Isolate surprisal contribution.
- Test whether compact features preserve performance.
- Quantify grounding-overlap trade-offs under identical prompting.

Quantitative Results (Salience Method)

Metric	Full	No Surprisal	Top-10
ROUGE-L	0.3488	0.3600	0.3626
BERTScore	0.9052	0.9077	0.9076
QA-EM	0.5367	0.5127	0.5160
QA-Consistency-F1	0.7752	0.7644	0.7718
Answer-Recovery-F1	0.4020	0.4302	0.4246

Main read:

- Full features lead on QA consistency and EM.
- No surprisal and Top-10 lead on lexical overlap and answer recovery.

Baseline Context (Per Run)

Metric	Full Run Baseline	No Surprisal Baseline	Top-10 Baseline
ROUGE-L	0.3504	0.3687	0.3710
BERTScore	0.9061	0.9094	0.9096
QA-EM	0.5687	0.5780	0.5613
QA-Consistency-F1	0.7670	0.7776	0.7653
Answer-Recovery-F1	0.4148	0.4537	0.4472

Interpretation caution:

- Baseline remains strong on overlap metrics.
- Salience is not intended to maximize lexical match; it prioritizes grounded targeting.

In-Depth Issue Analysis

Issue volume (about 1500 generated questions per condition):

- Full features run: ~ 2302 issue flags
- No surprisal run: ~ 2251 issue flags
- Top-10 run: ~ 2208 issue flags

Common issue reasons in `qa_generation_issues.json`:

- Low lexical overlap with closest gold question
- QA model answer mismatch
- Weak alignment with gold answers
- Generated question targets a different fact than gold

Key nuance: many flagged cases are valid alternate facts from salient sentences, not true generation failures.

What the Issue Patterns Reveal

- Salience guidance often generates plausible questions anchored to different passage facts than the gold QA pair.
- This can lower ROUGE-L while preserving or improving consistency-oriented metrics.
- QA extraction mismatch is partly an evaluator bottleneck: generated questions may be valid but out-of-distribution for the QA model.

Takeaway: metric disagreement here is informative, not noise.

- Higher grounding can come with lower lexical overlap.
- Better overlap does not always mean better salience alignment.

Key Insights & Nuances

1. Full features are strongest for consistency-focused behavior.
2. No-surprisal and Top-10 are competitive and simpler.
3. The choice of metric encodes the research objective.
4. Issue logs show many "different-but-valid" generations.
5. Overall evidence supports salience guidance as promising, not definitive.

Methodological implication:

- Evaluate with a metric portfolio, not a single overlap score.

Limitations

- Single prompt family and model configuration in this ablation set.
- Dependence on one QA evaluator introduces evaluator bias.
- Automatic metrics cannot fully capture pedagogical value.
- Issue thresholds are heuristic and can over-flag valid outputs.
- Current salience is sentence-level; discourse-level salience is not modeled yet.

Primary extension: RST-based discourse salience

- Incorporate Rhetorical Structure Theory (RST) features.
- Model discourse relations and nucleus/satellite roles.
- Combine sentence-level and discourse-level salience for better targeting.

Additional steps

- Human evaluation of grounded usefulness and factuality.
- Multi-QA evaluator setup for robust answer consistency estimates.
- Larger-scale runs and calibration analysis.

Conclusion & Open Questions

Conclusion

- Saliency-guided QG is a viable framework with clear trade-offs.
- Full features favor consistency; reduced sets favor overlap/recovery.
- Issue analysis supports the grounding-versus-overlap interpretation.

Open questions

- Which objective matters most for your downstream use case?
- How should we weight consistency vs lexical overlap in model selection?
- Does RST reduce different-fact drift while preserving diversity?

Thank you.

Questions, feedback, and suggestions for the next iteration are very welcome.